

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **machine learning pipelines** using **Python (pandas, NumPy, scikit-learn)** to automate feature engineering and model training across **AWS Lambda** and **EC2**, reducing manual data processing time by **60%** and enabling real-time inference at scale.
- Developed **predictive models** for transaction anomaly detection in payments processing using **supervised learning** (classification, regression) and **unsupervised clustering**, achieving **94% accuracy** in identifying fraudulent SWIFT message flows and reducing false positives by **35%**.
- Built end-to-end **data pipelines** and **ETL workflows** using **Python** and **SQL** (PostgreSQL, Oracle, MySQL) to ingest, transform, and validate financial transaction data; automated scheduling via **cron jobs** and **Linux shell scripting** to achieve **99.8% uptime** for production analytics workflows.
- Engineered **AI-powered automation workflows** integrating **Claude API** and **Anthropic SDK** with structured **prompt engineering** and **JSON constraint guardrails** for intelligent content generation; orchestrated **GitHub Actions CI/CD** pipelines and **Vercel deployment** to deliver scalable, production-ready AI systems.
- Created interactive **BI dashboards** and **data visualisations** using **Power BI** (DAX measures, Power Query, data modelling) and **Tableau** to surface model insights and business KPIs for cross-functional stakeholders; automated report generation reducing manual reporting overhead by **45%**.
- Performed statistical analysis and **hypothesis testing** on financial datasets using **R** and **RStudio**; applied **time-series forecasting** and **A/B testing methodologies** to validate model performance and drive data-informed decision-making across payments operations and risk management teams.

SKILLS

Machine Learning & AI – supervised learning, unsupervised learning, classification, regression, clustering, anomaly detection, model evaluation, hyperparameter tuning, prompt engineering, Claude API, Anthropic SDK.

Data Science & Analytics – statistical analysis, hypothesis testing, A/B testing, time-series forecasting, exploratory data analysis, feature engineering, model validation, scikit-learn, pandas, NumPy, R, RStudio.

Data Engineering & Pipelines – ETL development, data pipeline orchestration, SQL (PostgreSQL, Oracle, MySQL), Python scripting, cron scheduling, Linux shell scripting, data validation, schema design.

Business Intelligence – Power BI, Tableau, DAX, Power Query, data modelling, interactive dashboards, data visualisation, report automation.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD, GitHub Actions, Git version control, deployment automation.

Backend & API Development – Java Spring Boot, REST APIs, FastAPI, Flask, application integration, microservices architecture.

Domain Expertise – payments processing, SWIFT messaging (MT/MX), ISO 20022 message flows, financial services, transaction monitoring, regulatory compliance, GDPR, SEPA, cross-border payments.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

AI-Driven Resume Generation & Job Matching System (Personal Project)

- Architected end-to-end **machine learning workflow** leveraging **Claude API** and **Anthropic SDK** with advanced **prompt engineering** techniques to generate JD-tailored resume content at scale using **claude-haiku-4-5** model.
- Implemented **structured JSON prompt engineering** with constraint guardrails and validation logic to ensure model output consistency and quality; orchestrated data flow from **Apify web scrapers** → job description ingestion → AI-powered content generation → **reportlab PDF generation**.
- Deployed production system using **GitHub Actions CI/CD** for automated testing and **Vercel** for scalable cloud hosting; achieved **sub-2-second** inference latency and processed **500+ job descriptions** monthly, demonstrating hands-on mastery of AI automation and system design.

SWIFT Payments Monitoring & Anomaly Detection Dashboard (HCL Technologies)

- Developed **machine learning anomaly detection models** using **scikit-learn** to identify suspicious SWIFT message flows and payment transaction patterns; trained **unsupervised clustering algorithms** on historical transaction data achieving **94% detection accuracy** and reducing operational response time by **40%**.
- Built real-time **monitoring dashboards** using **Splunk**, **Dynatrace**, and **CloudWatch** integrated with **Python-based data pipelines** to surface ML model predictions and operational alerts; automated **ETL workflows** via **SQL queries** (PostgreSQL, Oracle) to ingest and validate ISO 20022 message data for **SLA compliance** and regulatory audit trails.
- Engineered production support system supporting **P1/P2 incident response** with **99.8% uptime** for payments operations; collaborated cross-functionally with banking domain teams on **SWIFT messaging protocols**, **SEPA compliance**, and regulatory requirements including GDPR and transaction reconciliation workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)